

Software Requirements Specification

Project: Ticketing Support System

Team: Group 4

Introduction and Context

The Ticketing Support System is a web-based IT support and incident management application, enabling customers to report issues and submit change requests easily to provide the support staff with the tools they need to triage, assign, and resolve tickets effectively. The ultimate purpose of the MVP is to create a secure, role-based workflow so that there is transparency, an appropriate distribution of work, and full action logging with no extra operating overhead.

Functional Requirements

Functional Requirement 01: Customer Ticket Submission

User Story: As a Customer, I want to submit an incident or a Request for Change (RFC), so that I can get help with my issues.

Acceptance Criteria:

- The user must be successfully authenticated as a Customer.
- The ticket creation form must require the user to input a "Title", "Description", and select a ticket "Type" (Incident or RFC).
- Upon submission, the system must save the ticket to the database with an automatically assigned status of "Open" and the current timestamp.
- The Customer must be redirected to a confirmation view displaying their new Ticket ID.

Functional Requirement 02: Customer Ticket Overview

User Story: As a Customer, I want to view the status of my own tickets, so that I know the progress of my request.

Acceptance Criteria:

- The system must retrieve and display only the tickets created by the currently logged-in Customer.
- The overview must display the Ticket ID, Title, Status, and Last Updated timestamp for each ticket.
- Customers must not be able to view tickets created by other users or internal tickets created by staff.

Functional Requirement 03: Support Agent Triage

User Story: As a Support Agent, I want to accept or deny a new ticket, so that I can filter out invalid requests.

Acceptance Criteria:

- The Support Agent must have access to a dashboard displaying all unassigned tickets with the status "Open".
- The Agent must be able to click an "Accept" button to approve the ticket for processing.
- The Agent must be able to click a "Deny" button, which requires providing a mandatory text reason before changing the ticket status to "Closed/Denied".

Functional Requirement 04: Ticket Assignment

User Story: As a Support Agent, I want to assign a ticket to a specific consultant, so that the right expert can handle the issue.

Acceptance Criteria:

- For accepted tickets, the Agent must be able to select an active Consultant from a system dropdown menu.

- Upon assignment, the ticket's assigned_to field must be updated with the Consultant's ID, and the ticket status must change to "Assigned".

Functional Requirement 05: Consultant Internal Tickets

User Story: As a Consultant, I want to create internal tickets for actions I take myself, so that my work is documented and transparent.

Acceptance Criteria:

- The Consultant must be able to create a new ticket and mark it specifically with an "Internal" flag.
- Internal tickets must be visible to Support Agents, Consultants, and Support Managers, but strictly hidden from all Customers.

Functional Requirement 06: Ticket Resolution and Communication

User Story: As a Consultant, I want to add comments and change the status of my assigned tickets, so that the customer stays informed.

Acceptance Criteria:

- The Consultant must only be able to modify tickets where they are the assigned user.
- The Consultant must be able to change the ticket status (e.g., to "In Progress" or "Resolved").
- The Consultant must be able to add text comments to the ticket.
- When adding a comment, the Consultant must have the option to check an "Internal" flag; if checked, the comment must be hidden from the Customer view.

Functional Requirement 07: Manager Workload Overview

User Story: As a Support Manager, I want an overview of all open and closed tickets, so that I can monitor the team's workload.

Acceptance Criteria:

- The Manager must have access to a global dashboard displaying all tickets in the system regardless of the creator or assignee.
- The Manager must be able to filter the displayed tickets by Status (Open, Assigned, Resolved, Closed) and by Assigned Consultant.

Functional Requirement 08: Audit and Compliance Logging

User Story: As a Support Manager, I want to see a log of all actions (who, what, when), so that the system remains compliant and secure.

Acceptance Criteria:

- The system must automatically generate a log entry in the database whenever a ticket is created, assigned, or changes status.
- The log entry must capture the User ID performing the action, the specific Action Type, the Ticket ID, and the Timestamp.
- The Manager must be able to view a read-only table of these audit logs.

Non-Functional Requirements

NFR-01 (Authentication):

The system must secure user sessions using a custom, natively implemented JSON Web Token (JWT) architecture.

NFR-02 (Security):

All user passwords must be hashed using the BCrypt algorithm prior to being stored in the database.

NFR-03 (Resource Management):

The backend must utilize an asynchronous scheduled background task to automatically locate and delete expired session tokens from the database every hour to optimize performance.

NFR-04 (Architecture):

The application must be structured using standard Java Enterprise (Jakarta EE) principles, utilizing a clear separation of concerns (e.g., DAO layer, Model layer, Security Filters).

"Won't Have" Requirements

WH-01:

The system will not support the uploading, storing, or downloading of file attachments for tickets.

WH-02:

The system will not integrate with third-party email or SMS providers to send automated external notifications upon ticket updates.

WH-03:

The system will not include single sign-on (SSO) integrations with external providers (such as Google or Microsoft).